$$e_i = y_i - f(x_i)$$

Consider $S = [y_1 - f(x_1)]^2 + [y_2 - f(x_2)]^2 + ...[y_m - f(x_m)]^2$

$$= e_1^2 + e_2^2 + ... + e_m^2$$

The method of least squares consists of minimising S.

5.2 FITTING A STRAIGHT LINE

Let $y = a_0 + a_1 x$ is a straight line to be fitted to the given data. Then

$S = [y_1 - (a_0 + a_1 x_1)]^2 + [y_2 - (a_0 + a_1 x_2)]^2 + ... + [y_m - (a_0 + a_1 x_m)]^2$

If S is to be minimum, we must have

$$\frac{\partial S}{\partial a_0} = 0 \Rightarrow -2[y_1 - (a_0 + a_1 x_1)] - 2[y_2 - (a_0 + a_1 x_2)] - ....$$

$$- 2[y_m - (a_0 + a_1 x_m)] = 0$$

$$\frac{\partial S}{\partial a_1} = 0 \Rightarrow -2x_1[y_1 - (a_0 + a_1 x_1)] - 2x_2[y_2 - (a_0 + a_1 x_2)] - ....$$

$$- 2x_m[y_m - (a_0 + a_1 x_m)] = 0$$

iQOO Z6 Lite
Feb 26, 2026, 11:26

Since we know x_i and y_i, equations (4) and (5) can be so
a_0 and a_1. Equations (7) and (8) are called *normal equations.*
These equations can be written as

$$\boxed{\begin{aligned} \Sigma y &= m a_0 + a_1 \Sigma x \\ \Sigma xy &= a_0 \Sigma x + a_1 \Sigma x^2 \end{aligned}}$$

Differentiating (4) and (5) once again w.r.t. a_0 and a_1 we

We find that these two will be positive at a_0 and a_1.

iQOO Z6 Lite
Feb 26, 2026, 11:26

$$= 15 - 38.769 = -23.7692$$

$\Rightarrow a = -3.9615$

$\therefore$ The required line is, $y = (-3.9615) + (2.1538)x$

Example 15 : The following are data on the drying time of a certain paint and the amount of an additive that is intended to reduce the drying time

Amount of paint additive (grams) x	0	1	2	3	4	5	6
Drying time (hrs) y	12	10.5	10	8	7	8	7.5

Fit a straight line $y = a + bx$, by the method of least squares.

[JNTU (K) II Sem. June 2015 (Set No. 4)]

iQOO Z6 Lite
Feb 26, 2026, 11:29

Solution : We write the following table

x	y	x^2	xy
0	12	0	0
1	10.5	1	10.5
2	10	4	20
3	8	9	24
4	7	16	28
5	8	25	40
6	7.5	36	45
$\Sigma x = 21$	$\Sigma y = 63$	$\Sigma x^2 = 91$	$\Sigma xy = 167.5$

The normal equations are

Solution : Given values are tab

x
20
60
100
140
180
220
260
300
340
380
2000

iQOO Z6 Lite
Feb 26, 2026, 11:29

$$\Sigma y = na_0 + a_1 \Sigma x$$

$$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2$$

submit the values.

$$63 = 7a_0 + a_1(21)$$

$$167.5 = a_0(21) + a_1(91)$$

simplification

$$7a_0 + 21a_1 = 63$$

$$21a_0 + 91a_1 = 167.5$$

solve the equations

$$a_0 = 11.30, \quad a_1 = -0.7678$$

Now

$$y = a_0 + a_1 x$$

$$y = 11.30 - 0.7678x$$

$55a_1 + 15a_0 = 748$

Solving, we get $a_0 = 0$, and $a_1 = 13.6$.

Thus the line of best fitting is $y = 13.6x$.

Example 2 : The temperatures T (in °c) and lengths L (in mm) of a heated rod are given below. If $L = a_0 + a_1 T$, find the best values for a_0 and a_1.

T	20	30	40	50	60	70
L	800.3	800.4	800.6	800.7	800.9	801.0

iQOO Z6 Lite
Feb 26, 2026, 11:27

x	y	x^2	xy
20	800.3	400	16006
30	800.4	900	24012
40	800.6	1600	32024
50	800.7	2500	40035
60	800.9	3600	48054
70	801.0	4900	56070
$\Sigma x = 270$	$\Sigma y = 4803.9$	$\Sigma x^2 = 13900$	$\Sigma xy = 216201$

Normal eq: $\Sigma y = m a_0 + a_1 \Sigma x$

$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2$

Submit values.

$4803.9 = 6 a_0 + a_1 (270)$

$216201 = a_0 (270) + a_1 (13900)$

simplification and order

$6 a_0 + 270 a_1 = 4803.9$

$270 a_0 + 13900 a_1 = 216201$

solving

on solving

$a_0 = 800$

$a_1 = 0.0146$

$y = a_0 + a_1 x$

$y = 800 + 0.0146 x$

70

$\Sigma x = 270$

Second degree polynomial or parabola curve is Given by

$$y = a + bx + cx^2$$

Normal eqs are

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

$$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

fit a second degree polynomial for the following data by the method of least squares.

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

Sol: second degree polynomial is $y = a + bx + cx^2$

Normal eqs: $\Sigma y = na + b\Sigma x + c\Sigma x^2$

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$

$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$

x	y	x^2	x^3	x^4	xy	x^2y
0	1	0	0	0	0	0
1	1.8	1	1	1	1.8	1.8
2	1.3	4	8	16	2.6	5.2
3	2.5	9	27	81	7.5	22.5
4	6.3	16	64	256	25.2	100.8
$\Sigma x = 10$	$\Sigma y = 12.9$	$\Sigma x^2 = 30$	$\Sigma x^3 = 100$	$\Sigma x^4 = 354$	$\Sigma xy = 37.1$	$\Sigma x^2y = 130.3$

submit in Normal eqs

$12.9 = 5a + b(10) + c(30) \Rightarrow 5a + 10b + 30c = 12.9$

$37.1 = a(10) + b(30) + c(100) \Rightarrow 10a + 30b + 100c = 37.1$

$130.3 = a(30) + b(100) + c(354) \Rightarrow 30a + 100b + 354c = 130.3$

on solving $a = 1.42$ $b = -1.07$ $c = 0.55$

submit in $y = a + bx + cx^2 \Rightarrow y = 1.42 - 1.07x + 0.55x^2$

Fit the least square parabola $y = a + bx + cx^2$ to the data $f(-1) = -2$, $f(0) = 1$, $f(1) = 2$, $f(2) = 4$

Sol:

x	-1	0	1	2
y	-2	1	2	4

Parabola eq is $y = a + bx + cx^2$

Normal eqs are $\Sigma y = ma + b\Sigma x + c\Sigma x^2$

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$

$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$

x	y	x^2	x^3	x^4	xy	x^2y
-1	-2	1	-1	1	2	-2
0	1	0	0	0	0	0
1	2	1	1	1	2	2
2	4	4	8	16	8	16
$\Sigma x = 2$	$\Sigma y = 5$	$\Sigma x^2 = 6$	$\Sigma x^3 = 8$	$\Sigma x^4 = 18$	$\Sigma xy = 12$	$\Sigma x^2 y = 16$

Substitute in normal eqs

$5 = 4a + b(2) + c(6) \Rightarrow 4a + 2b + 6c = 5$

$12 = a(2) + b(6) + c(8) \Rightarrow 2a + 6b + 8c = 12$

$16 = a(6) + b(8) + c(18) \Rightarrow 6a + 8b + 18c = 16$

Solve the eqs $a = 0.55$, $b = 2.15$ $c = -0.25$

Required parabola $y = a + bx + cx^2$

$= 0.55 + 2.15x - 0.25x^2$

Fond the parabola of the form $y = a + bx + cx^2$ passing through the points $(-1, 2)$ $(0, 1)$ and $(1, 4)$

Ans:

x	-1	0	1
y	2	1	4

x	y	x^2	x^3	x^4	xy	x^2y
-1	2	1	-1	1	-2	2
0	1	0	0	0	0	0
1	4	1	1	1	4	4
$\Sigma x = 0$	$\Sigma y = 7$	$\Sigma x^2 = 2$	$\Sigma x^3 = 0$	$\Sigma x^4 = 2$	$\Sigma xy = 2$	$\Sigma x^2y = 6$

Normal eqs are

$7 = 3a + b(0) + c(2) \Rightarrow \quad 3a + 0b + 2c = 7$

$2 = a(0) + b(2) + c(0) \Rightarrow \quad 0a + 2b + 0c = 2$

$6 = a(2) + b(0) + c(2) \Rightarrow \quad 2a + 0b + 2c = 6$

on solving we get

$a = 1, \quad b = 1, \quad c = 2$

req parabola is $y = a + bx + cx^2$

$y = 1 + x + 2x^2$

Fitting exponential curve of the form

$$y = a e^{bx}$$

the Normal eqs are

~~$\Sigma \log y = m \log a$~~

$$\Sigma \ln y = m \ln a + b \Sigma x$$

$$\Sigma x \ln y = \ln a \, \Sigma x + b \Sigma x^2$$

Find a curve $y = ae^{bx}$ to the data

x	0	2	4
y	5.1	10	31.1

curve: $y = ae^{bx}$

Normal eqs: $\Sigma \ln y = n \ln a + b \Sigma x$

$\Sigma x \ln y = \ln a \, \Sigma x + b \, \Sigma x^2$

x	y	$\ln y$	x^2	$x \ln y$
0	5.1	1.6292	0	0
2	10	2.3026	4	4.6050
4	31.1	3.4372	16	13.7486
$\Sigma x = 6$		$\Sigma \ln y = 7.3689$	$\Sigma x^2 = 20$	$\Sigma x \ln y = 18.3538$

$3 \ln a + 6b = 7.3689$

$6 \ln a + 20b = 18.3538$

on solving $\ln a = 1.5523$, $b = 0.4520$

$e^{\ln a} = e^{1.5523}$

$a = 4.7223$

$y = ae^{bx} \Rightarrow y = 4.7223 \, e^{0.4520x}$

Determine the constants a and b by the method of least squares so that $y = ae^{bx}$

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

$\therefore$ exp curve $y = ae^{bx}$

Normal eqs are $\Sigma \ln y = n \ln a + b \Sigma x$

$\Sigma x \ln y = \ln a \, \Sigma x + b \Sigma x^2$

x	y	ln y	x^2	x ln y
2	4.077	1.405	4	2.810
4	11.084	2.405	16	9.620
6	30.128	3.405	36	20.430
8	81.897	4.405	64	35.240
10	222.62	5.405	100	54.050
$\Sigma x = 30$		$\Sigma \ln y = 17.025$	$\Sigma x^2 = 220$	$\Sigma x \ln y = 122.150$

Normal eqs $\Rightarrow$ $5 \ln a + 30b = 17.025$

$30 \ln a + 220b = 122.150$

on solving $\ln a = 0.4050 \quad b = 0.5$

$e^{\ln a} = e^{0.4050}$

$a = 1.499$

$\therefore y = ae^{bx} \Rightarrow y = 1.499 e^{0.5x}$

Curve $y = ab^x$

Normal eqs $\Sigma \log y = m \log a + \log b \, \Sigma x$

$\Sigma x \log y = \log a \, \Sigma x + \log b \, \Sigma x^2$

obtain a Relation of the from $y = ab^x$ for the following data

x	2	3	4	5	6
y	8.3	15.4	33.1	65.2	127.4

Curve : $y = ab^x$

Normal eqs: $\Sigma \log y = n \log a + \log b \, \Sigma x$

$\Sigma x \log y = \log a \, \Sigma x + \log b \, \Sigma x^2$

x	y	$\log y$	x^2	$x \log y$
2	8.3	0.9191	4	1.8382
3	15.4	1.1875	9	3.5625
4	33.1	1.5198	16	6.0792
5	65.2	1.8142	25	9.0710
6	127.4	2.1052	36	12.631
$\Sigma x = 20$		$\Sigma \log y = 7.5455$	$\Sigma x^2 = 90$	$\Sigma x \log y = 33.1819$

Submit values in normal eqs

$5 \log a + 20 \log b = 7.455$

$20 \log a + 90 \log b = 33.1819$

$\therefore \log a = 0.3091 \quad \log b = 0.3$

$10^{\log a} = 10^{0.3091} \quad 10^{\log b} = 10^{0.3}$

$a = 2.0375 \quad b = 1.9953$

$\therefore y = (2.0375)(1.9953)^x$

Fit $y = ab^x$ by the method of least squares

x	0	1	2	3	4	5	6	7
y	10	21	35	59	92	200	400	610

x	y	$\log y$	x^2	$x \log y$
0	10	1	0	0
1	21	1.3222	1	1.3222
2	35	1.5441	4	3.0882
3	59	1.7708	9	5.3124
4	92	1.9638	16	7.8552
5	200	2.3010	25	11.5050
6	400	2.6021	36	15.6126
7	610	2.7853	49	19.4971
$\Sigma x = 28$		$\Sigma \log y = 15.2893$	$\Sigma x^2 = 140$	$\Sigma x \log y = 64.1927$

Normal eqs: $8 \log a + 28 \log b = 15.2893$

$28 \log a + 140 \log b = 64.1927$

On solving $\log a = 1.0212$, $\log b = 0.2543$

$10^{\log a} = 10^{1.0212}$ $\quad 10^{\log b} = 10^{0.2543}$

$a = 10.5003$, $\quad b = 1.7960$

$y = (10.5003)(1.7960)^x$

power Curve ⇒ Geometric Curve

$$y = ax^b$$

$$\Sigma \ln y = m \ln a + b \Sigma \ln x$$

$$\Sigma \ln x \ln y = \ln a \Sigma \ln x + b \Sigma (\ln x)^2$$

Fit a power curve $y = ax^b$ for the following data.

x	2	3	4	5	6
y	8.3	15.4	33.1	65.2	127.4

Sol: Curve $y = ax^b$

Normal eqs: $\Sigma \ln y = n \ln a + b \Sigma \ln x$

$\Sigma \ln x \ln y = \ln a \Sigma \ln x + b \Sigma (\ln x)^2$

x	y	$\ln x$	$\ln y$	$(\ln x)^2$	$\ln x \ln y$
2	8.3	0.6931	2.1163	0.4804	1.4669
3	15.4	1.0986	2.7344	1.2069	3.0040
4	33.1	1.3863	3.4995	1.9218	4.8514
5	65.2	1.6094	4.1775	2.5902	6.7234
6	127.4	1.7918	4.8473	3.2104	8.6853
		$\Sigma \ln x = 6.5793$	$\Sigma \ln y = 17.3750$	$\Sigma (\ln x)^2 = 9.4097$	

$\Sigma \ln x \Sigma \ln y = 24.7310$

Normal eqs: $5 \ln a + 6.5793\, b = 17.3750$

$6.5793 \ln a + 9.4097\, b = 24.7310$

$\ln a = 0.2076 \qquad b = 2.4831$

$e^{\ln a} = e^{0.2076}$

$a = 1.2307$

$\therefore$ Curve $y = ax^b \Rightarrow y = 1.2307\, x^{2.4831}$

Fit a power function for the following Data for the demand of a product against its price ~~charge~~ change rate in five different cities. predict the demand when the price of the product is 12 Rs.

price	20	16	10	11	14
demand	22	41	120	89	56

Curve $y = ax^b$

Normal eq's $\Sigma \ln y = n \ln a + b \Sigma \ln x$

$\Sigma \ln x \ln y = \ln a \Sigma \ln x + b \Sigma (\ln x)^2$

$\ln x$	$\ln y$	$(\ln x)^2$	$\ln x \cdot \ln y$
2.9957	3.0910	8.9742	9.2597
2.7726	3.7136	7.6873	10.2963
2.3026	4.7875	5.3020	11.0237
2.3979	4.4886	5.7499	10.7632
2.6391	4.0254	6.9648	10.6234
$\Sigma \ln x = 13.1079$	$\Sigma \ln y = 20.1061$	$\Sigma (\ln x)^2 = 34.6782$	$\Sigma \ln x \ln y = 51.9663$

$5 \ln a + 13.1079\, b = 20.1061$

$13.1079 \ln a + 34.6782\, b = 51.9663$

$\ln a = 10.2127, \quad b = -2.3617$

$a = 27247.0354,$

$y = (27247.0354)\, x^{-2.3617}$

put $x = 12$

$y = 77.0224$

By paper transformation, Convert the relation $y = a + bxy$ to a linear form and find the equation to fit the data.

x	-4	1	2	3
y	4	6	10	8

Compare $y = a + bxy$ with $y = a_0 + a_1 x$

$\therefore a_0 = a, \quad a_1 = b, \quad x = xy$

Normal eqs of $y = a_0 + a_1 x$ are

~~$\Sigma y = a \cdot \Sigma x + b$~~

$$\Sigma y = na_0 + a_1 \Sigma x$$

$$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2$$

Replace $a_0 = a, \quad a_1 = b, \quad x = xy$

$$\Sigma y = na + b \Sigma xy$$

$$\Sigma xy^2 = a \Sigma xy + b \Sigma (xy)^2$$

x	y	xy	xy^2	$(xy)^2$
-4	4	-16	-64	256
1	6	6	36	36
2	10	20	200	400
3	8	24	192	576
	$\Sigma y = 28$	$\Sigma xy = 34$	$\Sigma xy^2 = 364$	$\Sigma (xy)^2 = 1268$

$$4a + 34b = 28$$

$$34a + 1268b = 364$$

$$a = 5.9060, \quad b = 0.1287$$

$$\therefore \quad y = 5.9060 + 0.1287x$$

Here a, b, h, k are constants (a is the assumed mean of X series and b is the assumed mean of Y series).

3.11 PROPERTIES OF CORRELATION COEFFICIENT

Property I : Show that the maximum value of rank correlation coefficient is 1.

(OR)

The coefficient of correlation lies between -1 and $+1$.

Symbolically, $-1 \leq r \leq +1$ or $|r| \leq 1$

Note : **1.** $-1 \leq r \leq 1$ *i.e.* correlation coefficient can not exceed one numerically.

2. If $r = 1$ correlation is perfect and positive.

3. If $r = -1$ correlation is perfect and negative.

4. If $r = 0$, then there is no relationship between the variables.

Property II :

iQOO Z6 Lite
Mar 2, 2026, 11:00

KARL PEARSON's Coefficient of Correlation

$$r = \frac{\text{Covariance of } (x,y)}{\sigma_x \sigma_y}$$

$$r = \frac{n \Sigma xy - \Sigma x \Sigma y}{\sqrt{n \Sigma x^2 - (\Sigma x)^2} \sqrt{n \Sigma y^2 - (\Sigma y)^2}}$$

$$r = \frac{\Sigma dx \cdot dy}{\sqrt{\Sigma dx^2} \sqrt{\Sigma dy^2}} \quad \text{or} \quad \frac{\Sigma XY}{\sqrt{\Sigma X^2 \cdot \Sigma Y^2}}$$

Calculate coefficient of correlation for the following data:

x	12	9	8	10	11	13	7
y	14	8	6	9	11	12	3

x	y	x^2	y^2	xy
12	14	144	196	168
9	8	81	64	72
8	6	64	36	48
10	9	100	81	90
11	11	121	121	121
13	12	169	144	156
7	3	49	9	21
$\Sigma x = 70$	$\Sigma y = 63$	$\Sigma x^2 = 728$	$\Sigma y^2 = 651$	$\Sigma xy = 676$

$$r = \frac{\text{Covariance of } (x, y)}{\sigma_x \sigma_y}$$

$$= \frac{n \Sigma xy - \Sigma x \Sigma y}{\sqrt{n \Sigma x^2 - (\Sigma x)^2} \sqrt{n \Sigma y^2 - (\Sigma y)^2}}$$

$$= \frac{7 \times 676 - 70 \times 63}{\sqrt{7 \times 728 - (70)^2} \sqrt{7 \times 651 - (63)^2}}$$

$r = 0.95$ (positive correlation)

Psychological tests of intelligence and of engineering ability were applied to 10 students. Here is a record of ungrouped data showing intelligence ration (I.R) and Engineering ratio (E.R). Find is there any significant correlation between the I.R and E.R

Student	A	B	C	D	E	F	G	H	I	J
IR	105	104	102	101	100	99	98	96	93	92
ER	101	103	100	98	95	96	104	92	97	94

Find mean $\bar{x}$ for x values

$$\bar{x} = \frac{105+104+102+101+100+99+98+96+93+92}{10}$$

$$\bar{x} = 99$$

Find mean $\bar{y}$ for y - values

$$\bar{y} = \frac{101+103+100+98+95+96+104+92+97+94}{10}$$

$$\bar{y} = \frac{980}{10} = 98$$

$$\bar{y} = 98$$

Let $dx = x - \bar{x}$, $dy = y - \bar{y}$

or $X = x - \bar{x}$ $Y = y - \bar{y}$

x	y	dx	dy	dx^2	dy^2	dx dy
105	101	6	3	36	9	18
104	103	5	5	25	25	25
102	100	3	2	9	4	6
101	98	2	0	4	0	0
100	95	1	-3	1	9	-3
99	96	0	-2	0	4	0
98	104	-1	6	1	36	-6
96	92	-3	-6	9	36	18
93	97	-6	-1	36	1	6
92	94	-7	-4	49	16	28
990	980	$\Sigma dx = 0$	$\Sigma dy = 0$		$\Sigma dy^2 = 140$	$\Sigma dx dy = 92$

$\Sigma dx^2 = 170$ $\Sigma dy^2 = 140$

Karl pearson Coefficient of Correlation

$$r = \frac{\Sigma dx\, dy}{\sqrt{\Sigma dx^2}\sqrt{\Sigma dy^2}} = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \Sigma y^2}}$$

$$= \frac{92}{\sqrt{170}\sqrt{140}}$$

$$r = 0.59$$

Correlation with Assumed mean

$$r = \frac{n \Sigma XY - \Sigma X \Sigma Y}{\sqrt{n \Sigma X^2 - (\Sigma X)^2}\ \sqrt{n \Sigma Y^2 - (\Sigma Y)^2}}$$

Calculate karl pearson correlation coefficient for the following paired data.

x	28	41	40	38	35	33	40	32	36	33
y	23	34	33	34	30	26	28	31	36	38

Sol:

Here ~~n=10~~ $n = 10$

Choose $\bar{x}$ from x-values

for example $\bar{x} = 35$

Choose $\bar{y}$ from y-values

for example $\bar{y} = 31$

x	y	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
28	23	-7	-8	49	64	56
41	34	6	3	36	9	18
40	33	5	2	25	4	10
38	34	3	3	9	9	9
35	30	0	-1	0	1	0
33	26	-2	-5	4	25	10
40	28	5	-3	25	9	-15
32	31	-3	0	9	0	0
35	36	1	5	1	25	5
33	38	-2	7	4	49	-4
		$\Sigma X = 6$	$\Sigma Y = 3$	$\Sigma X^2 = 162$	$\Sigma Y^2 = 195$	$\Sigma XY = 79$

$$r = \frac{n \Sigma XY - \Sigma X \Sigma Y}{\sqrt{n \Sigma X^2 - (\Sigma X)^2}\ \sqrt{n \Sigma Y^2 - (\Sigma Y)^2}}$$

$$= \frac{10(79) - 6 \times 3}{\sqrt{10 \times 162 - (6)^2}\ \sqrt{10 \times 195 - (3)^2}}$$

$$= 0.4403$$

Spearman's Rank Correlation

$$\rho = 1 - \frac{6\left[\Sigma D^2 + \frac{1}{2} m_1(m_1^2-1) + \frac{1}{2} m_2(m_2^2-1) + \frac{1}{2} m_3(m_3^2-1) + \cdots\right]}{n(n^2-1)}$$

Find rank Correlation for the following Data

maths marks	95	99	98	85	84
stat	89	75	88	79	92

x	y	Rx	Ry	D = Rx - Ry	D^2
95	89	3	2	1	1
99	75	1	5	-4	16
98	88	2	3	-1	1
85	79	4	4	0	0
84	92	5	1	4	16
					$\Sigma D^2 = 34$

$n = 5$

$$\rho = 1 - \frac{6 \Sigma D^2}{n(n^2-1)}$$

$$= 1 - \frac{6 \times 34}{5(5^2-1)}$$

$$= -0.7$$

Find Coefficient of Correlation between Industrial production and export using the following data

production	55	56	58	59	60	60	42
Export	35	38	38	39	44	43	45

x	R_x	y	R_y	$D = R_x - R_y$	D^2
55	6	35	7	-1	1
56	5	38	5.5	-0.5	0.25
58	4	38	5.5	-1.5	2.25
59	3	39	4	-1	1
60	1.5	44	2	-0.5	0.25
60	1.5	43	3	-1.5	2.25
42	7	45	1	6	36
					$\Sigma D^2 = 43$

$n = 7$,

For 60, $m_1 = 2$

For 38 $m_2 = 2$

$$\gamma = 1 - \frac{6\sum d^2 + \frac{1}{2} m_1(m_1^2-1) + \frac{1}{2} m_2(m_2^2-1) + \frac{1}{3} m_3(m_3^2-1) + \cdots}{n(n^2-1)}$$

$$= 1 - \frac{6(43) + \frac{1}{2}(2)(2^2-1) + \frac{1}{2}(2)(2^2-1)}{7(7^2-1)}$$

$$= 1 - \frac{258 + 3 + 3}{336}$$

$$= 0.2143$$

Find the rank correlation coefficient for the following Data

x	66	65	64	66	61	65	66
y	65	64	65	61	62	69	68

Sol:

x	R_x	y	R_y	$D = R_x - R_y$	D^2
66	2	65	3.5	-1.5	2.25
65	4.5	64	5	-0.5	0.25
64	6	65	3.5	2.5	6.25
66	2	61	7	-5	25
61	7	62	6	1	1
65	4.5	69	1	3.5	12.25
66	2	68	2	0	0

$\Sigma D^2 = 47$

$n = 7$.

For 66, $m_1 = 3$

For 65 $m_2 = 2$

For 65 $m_3 = 2$

Spearman's Rank Correlation

$$\rho = 1 - \frac{6\left[\sum D^2 + \frac{1}{2}m_1(m_1^2-1) + \frac{1}{2}m_2(m_2^2-1) + \frac{1}{2}m_3(m_3^2-1) + \cdots\right]}{n(n^2-1)}$$

$$\rho = 1 - \frac{6\left[(47) + \frac{1}{2}3(3^2-1) + \frac{1}{2}(2)(2^2-1) + \frac{1}{2}2(2^2-1)\right]}{7(7^2-1)}$$

$$\rho = 0.1071$$